Soft Robotic Grippers using Tactile Sensor

Report submitted to

Indian Institute of Technology, Kharagpur

For the fulfillment of BTP-I in

Mechanical Engineering with specialisation in
Manufacturing Science and Engineering by

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Certificate of Approval

This is to certify that the project report entitled **"Development, Non-linear Modeling, and Embedded Tactile Control of a 3D-Printed Compliant Soft Robotic Gripper"** submitted by **G. Chandana (23ME31008)**, **Shinde Sanchali Sanjay (23ME31029)**, and **Saurav Shekhar (23ME31043)** to the Indian Institute of Technology Kharagpur, for the partial fulfillment of the requirements for the award of the degree of *Bachelor of Technology in Mechanical Engineering (with specialization in Manufacturing Science and Engineering)*, is a bona fide record of the work carried out by them under my supervision and guidance.

The project is currently at the literature-review-completed stage, and the proposed methodology, numerical simulation framework, and experimental implementation roadmap have been thoroughly examined and approved for progression into Project Stage-II (BTP-II).

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Abstract

Traditional robotic manipulation systems rely on rigid articulated linkages that exhibit high positioning repeatability but lack intrinsic compliance, rendering them prone to crushing delicate, fragile, or irregularly contoured objects in automated agriculture, biomedical packaging, and explosive ordnance handling. Soft robotic end-effectors address this limitation through structural deformability, enabling passive geometric conformity and safe contact pressure distribution. However, conventional soft grippers suffer from severe engineering bottlenecks: low payload capacity, material hysteresis, complex tethered fluidic supplies, sensor-skin delamination, and an absence of localized feedback capable of detecting incipient slip without bulky optical apparati.

This report establishes the complete Stage-I research foundation and detailed Stage-II implementation roadmap for a **Monolithic 3D-Printed (FDM TPU) Tendon-Driven Compliant Soft Robotic Gripper with Embedded Tactile Sensing**. Synthesizing an extensive multi-author literature corpus, the research evaluates fluidic elastomer actuators (PneuNets), shape memory alloys (Nitinol), and cable-driven compliant architectures. To overcome mold-tooling delays and fluid leakage risks, a direct additive manufacturing route using Thermoplastic Polyurethane (TPU 95A) is formulated, integrating circular flexure living hinges that eliminate mechanical joint backlash while maximizing structural fatigue life.

To provide deterministic grasp autonomy, thin-film Force Sensitive Resistors (FSR402) are integrated flush into the phalange contact pads. An integrated mechatronic architecture centered on a dual-core ESP32 executing FreeRTOS is developed. The tactile signal pipeline decouples static normal contact force estimation (via an active second-order low-pass Butterworth filter at $f_c = 20$ Hz) from dynamic incipient slip micro-vibrations (via a 256-point Fast Fourier Transform analyzing the $100 - 300$ Hz spectral energy band). This enables closed-loop servomotor torque compensation in under 35 milliseconds. A non-linear finite element analysis (FEA) framework utilizing Mooney-Rivlin hyperelastic strain energy formulations is established to predict tip deflection and stress concentrations prior to physical fabrication. Finally, a structured 16-week semester roadmap is detailed, establishing quantitative benchmarks for pull-out retention force, payload-to-weight ratio, and an autonomous slip-prevention demonstration on physical hardware.

Keywords: Soft Robotics, Compliant Mechanisms, Additive Manufacturing, TPU 95A, Tactile Sensing, FSR402, Fast Fourier Transform, Incipient Slip Detection, FreeRTOS, Hyperelastic FEA.

Chapter 1: Introduction

1.1 Background of Soft Robotics and Compliant Manipulation

Robotic manipulation systems have traditionally been engineered around rigid mechanical kinematic chains composed of metallic or hard polymeric links joined by discrete revolute, prismatic, or spherical hinges. Such systems achieve sub-millimeter positioning accuracy, high structural rigidity, and substantial load-bearing capacities when operating in structured industrial environments where target workpieces possess invariant dimensions, known orientations, and high mechanical stiffness.

However, modern automation requirements are rapidly shifting toward unstructured and semi-structured domains, such as the selective harvesting of fragile agricultural produce (e.g., tomatoes, strawberries, peaches), handling delicate specimens in biomedical logistics, sorting irregular post-consumer recycling items, and conducting hazardous explosive ordnance disposal (EOD). In these settings, traditional rigid end-effectors exhibit intrinsic limitations: localized stress concentrations causing workpiece rupture, severe control complexity for sensorless grasping of unknown contours, and dynamic contact chatter leading to dropped objects.

Soft robotics represents a paradigm shift in end-effector design by replacing rigid links and discrete mechanical pins with continuously deformable elastomeric bodies and monolithic compliant mechanisms. By embodying mechanical intelligence directly into the elasticity of the material, soft grippers passively adapt their morphological shape to envelop complex contours upon contact, distributing interaction forces over an expanded surface area.

1.2 Motivation and Industrial Relevance

The primary motivation for this project stems from the critical need for a low-cost, self-contained, lightweight, and sensorized soft robotic gripper capable of executing delicate manipulation while providing real-time feedback regarding contact force and grasp stability.

Specific target applications include:

- **Automated Agricultural Harvesting:** Gentle detachment and transfer of soft-skinned fruits and vegetables without inducing bruised tissue that accelerates biological decay.
- **Biomedical and Pharmaceutical Logistics:** Sorting and transferring thin-walled glass vials, filled fluid ampoules, surgical specimens, and laboratory glassware where mechanical shock must be strictly eliminated.
- **Explosive Ordnance Disposal (EOD):** Interfacing with uncertain, aged, or structurally compromised explosive devices where uncontrolled impact, spark generation, or excessive clamping force could trigger catastrophic detonation.

1.3 Research Issues and Technical Challenges

While passive compliance provides adaptability, it introduces substantial engineering challenges that do not exist in rigid mechanisms:

1. **Non-linear Continuum Mechanics:** Soft elastomeric structures undergo geometric and material non-linearities requiring hyperelastic constitutive formulations.
2. **Hysteresis and Viscoelastic Relaxation:** Energy dissipation causes loading and unloading curves to diverge, reducing open-loop repeatability.
3. **Tethering Bottlenecks:** Conventional pneumatic soft grippers demand heavy compressors, pumps, and valves, restricting mobile deployment.
4. **Sensory Blindness:** Without embedded tactile sensing, the controller cannot ascertain grasp stability or detect slip transitions.
5. **Interfacial Delamination:** Embedding planar sensors onto stretching elastomeric skins induces severe peeling shear stresses.

1.4 Broad Objectives

To overcome these interconnected challenges, this research develops a monolithic 3D-printed compliant finger architecture fabricated from Thermoplastic Polyurethane (TPU 95A), driven by remote base-mounted servomotors via high-strength Dyneema tendons, equipped with embedded FSR402 tactile sensors, and governed by a dual-core ESP32 running FreeRTOS and micro-ROS for autonomous slip detection.

Chapter 2: Literature Review

2.1 Synthesis of Available Literature Work

The literature review conducted by the team synthesizes three distinct thematic investigations:

- **Actuation Schemes and Compliant Mechanisms (Shinde Sanchali Sanjay):** Evaluated pneumatic, tendon-driven, SMA, and DEA architectures, emphasizing the trade-off between structural compliance and load capacity.
- **Materials Science and Additive Manufacturing (G. Chandana):** Analyzed silicones (Ecoflex, PDMS), TPUs, and flexible photopolymers, concluding that direct FDM 3D printing of TPU filament eliminates mold-making delays and chemical curing hazards.
- **Tactile Transduction and Feedback Technologies (Saurav Shekhar):** Examined piezoresistive, capacitive, piezoelectric, and optical sensors, establishing FSR402 thin-film piezoresistors as optimal for grasping force detection.
- **Joint Framework (BTP-I Joint Report):** Formulated the preliminary system uniting compliant structures, planned hyperelastic FEA, FreeRTOS embedded processing, and FFT-based slip detection.

2.2 Comparative Analysis of Existing Gripper Technologies

Metric / Parameter	Pneumatic PneuNets	Tendon-Driven TPU (Selected)	Shape Memory Alloy (SMA)	Dielectric Elastomer (DEA)
Actuation Physics	Fluidic cavity pressure (ΔP)	Mechanical cable tension (T)	Joule-heated phase change ($M \leftrightarrow A$)	Maxwell electrostatic stress ($\sigma = \epsilon E^2$)
Primary Material	Silicone (Ecoflex, PDMS)	TPU 95A / 85A filament	Polymer + Nitinol wire	Silicone film + carbon grease
Fabrication Route	Multi-part mold casting	Direct FDM 3D printing	Pre-straining & embedding	Multi-layer spin coating
Response Time	0.5 – 1.2 s	**< 0.15 s**	3.0 – 15.0 s (cooling lag)	< 0.05 s
Payload-to-Weight	Moderate (1:1 – 2:1)	**High (3:1 – 5:1)**	Low-Moderate (0.5:1)	Low (< 0.5:1)
System Portability	Low (Tethered pump/valves)	**Maximum (Self-contained)**	High (DC battery operated)	Low (Heavy high-voltage supply)
Control Simplicity	Moderate (Pneumatic valves)	**High (Standard servo PWM)**	Very Low (Severe hysteresis)	Moderate (High-voltage switching)

Table 2.1: Multi-parameter comparison of candidate soft robotic gripper technologies.

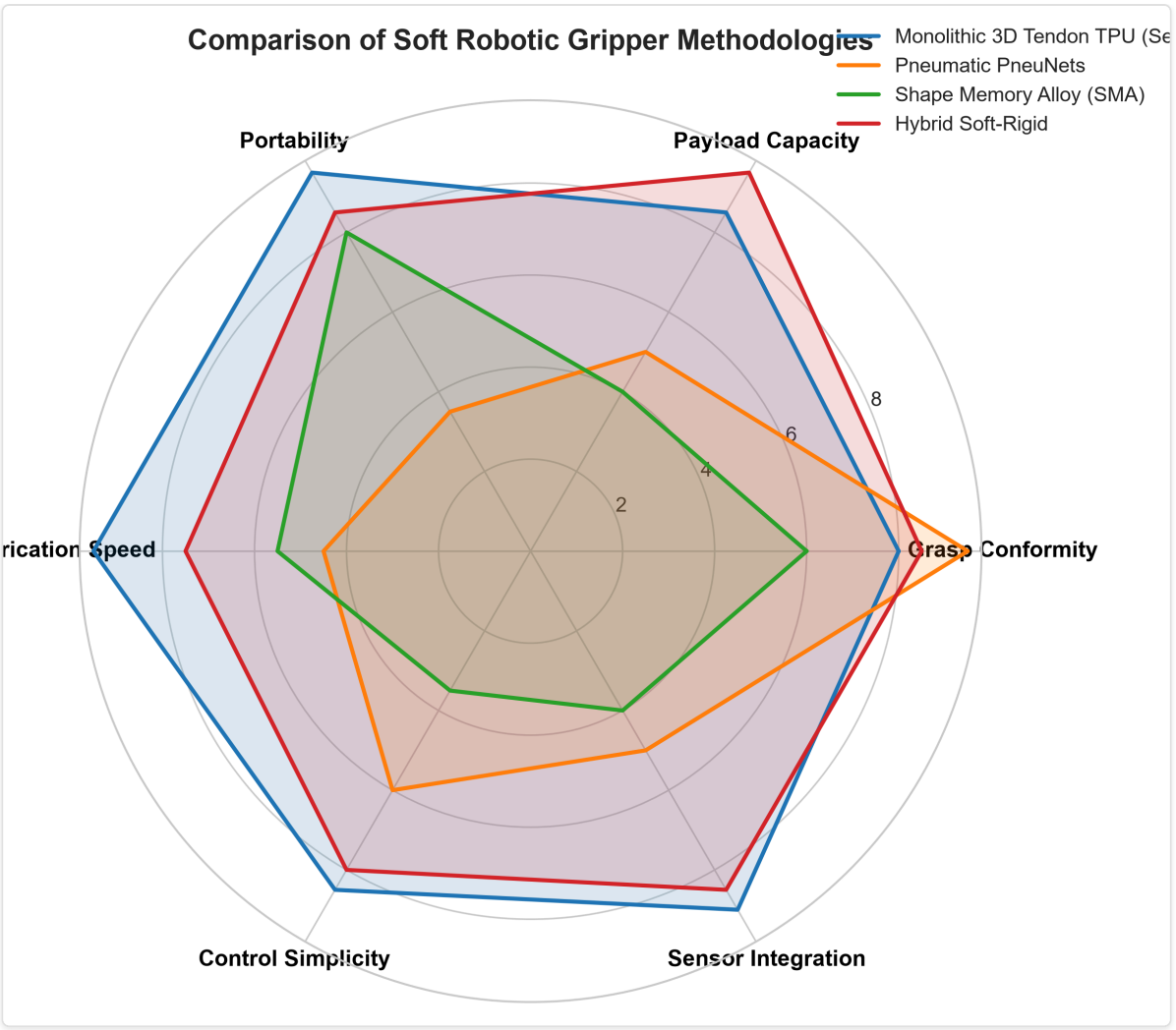


Figure 2.1: Multi-attribute radar comparison of candidate actuation technologies.

2.3 Research Gaps and Limitations

1. **Research Gap 1: Structural Compliance vs. Payload Capacity:** Soft bladders buckle under lateral loads; rigid-compliant hybrid flexures bridge this gap.
2. **Research Gap 2: Sensor Delamination under Cyclic Bending:** Planar sensors bonded to stretching elastomeric skins peel under cyclic deformation.
3. **Research Gap 3: Absence of Compact Incipient Slip Detection:** Existing slip-detecting grippers rely on heavy optical cameras (GelSight) unsuited for mobile arms.
4. **Research Gap 4: Material Viscoelastic Hysteresis:** Open-loop controllers ignore TPU relaxation during sustained grasps.
5. **Research Gap 5: Edge Control Disconnect:** Many prototypes lack deterministic onboard real-time operating systems integrated with ROS 2.

Chapter 3: Objectives and Scope of Work

3.1 Main Objective

To design, model, fabricate, and experimentally validate an integrated **Monolithic 3D-Printed Tendon-Driven Compliant Soft Robotic Gripper** featuring embedded piezoresistive tactile sensing and real-time edge processing on an ESP32 microcontroller, culminating in an autonomous closed-loop demonstration of incipient slip detection and force compensation.

3.2 Specific Sub-Objectives

1. Synthesize parametric 3D CAD models of compliant fingers with optimized circular living hinges and recessed sensor beds (planned for Stage-II).
2. Formulate and execute non-linear hyperelastic FEA in Ansys Mechanical (planned for Stage-II) to predict bending deflections and verify safety factors.
3. Optimize direct-drive FDM 3D printing of TPU 95A to ensure high inter-layer tensile strength.
4. Design and fabricate an active analog signal conditioning circuit (2nd-order Butterworth low-pass filter, $f_c = 20$ Hz).
5. Implement an on-chip 256-point FFT algorithm on ESP32 Core 0 to detect micro-vibrations ($100 - 300$ Hz) indicative of incipient slip.
6. Develop FreeRTOS dual-core firmware for closed-loop servo torque tightening in under 35 ms.
7. Quantify static force calibration, pull-out retention force, and payload-to-weight ratio across a standardized test library.

3.3 Scope and Limitations

- **In-Scope:** CAD modeling, non-linear hyperelastic FEA, FDM 3D printing of TPU/PLA, FSR402 sensor integration, FreeRTOS firmware, and static/dynamic pull-out testing.
- **Out-of-Scope:** High-speed industrial pick-and-place (> 3 m/s), multi-axis shear force sensor arrays, underwater manipulation, and explosive field trials.

Chapter 4: Proposed Methodology

4.1 System Architecture

The proposed system integrates mechanical compliance, tendon actuation, embedded tactile sensing, and dual-core edge processing into a cohesive mechatronic end-effector.

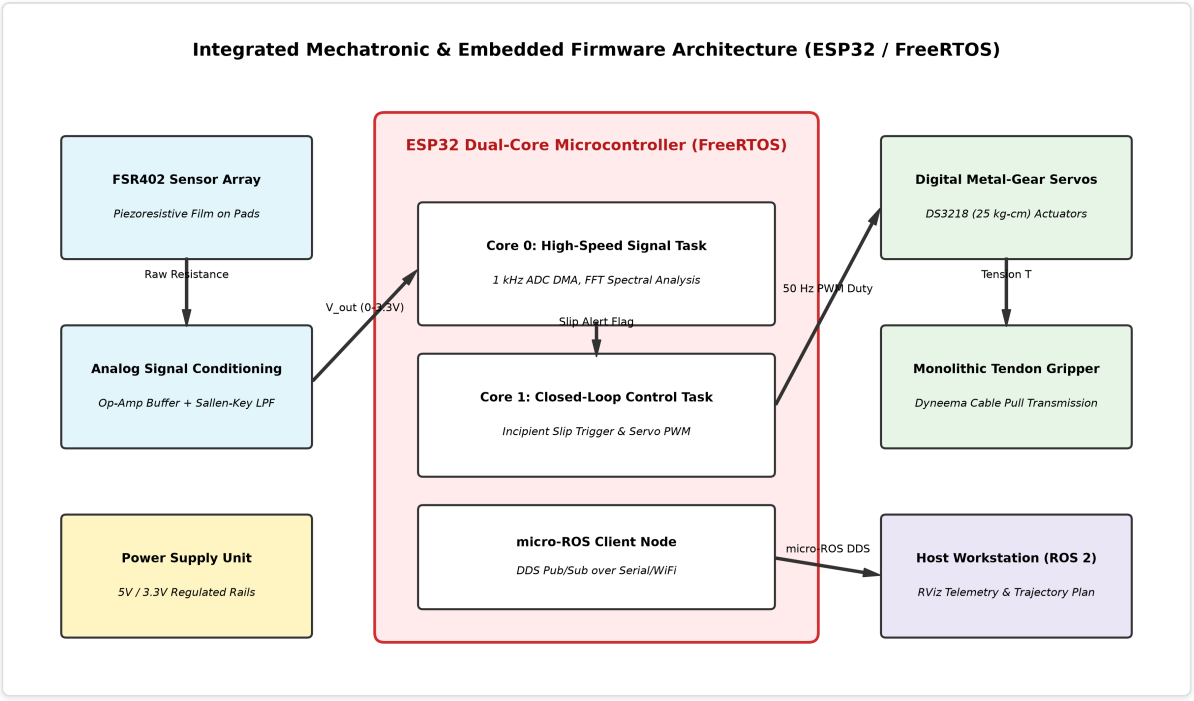


Figure 4.1: Integrated mechatronic and FreeRTOS embedded architecture for the compliant gripper.

4.2 Kinematic Modeling and Living Hinge Synthesis

The finger is modeled as a monolithic beam with three localized circular flexure notches replicating human MCP, PIP, and DIP joints. According to Howell's Pseudo-Rigid-Body Model (PRBM), the equivalent torsional spring stiffness K_θ is expressed as:

$$K_\theta = \frac{E \cdot w \cdot t^{2.5}}{9\pi\sqrt{r}}$$

where E is the instantaneous flexural modulus of TPU 95A, w is finger width, t is flexure throat thickness (1.6 mm), and r is notch radius (2.5 mm).

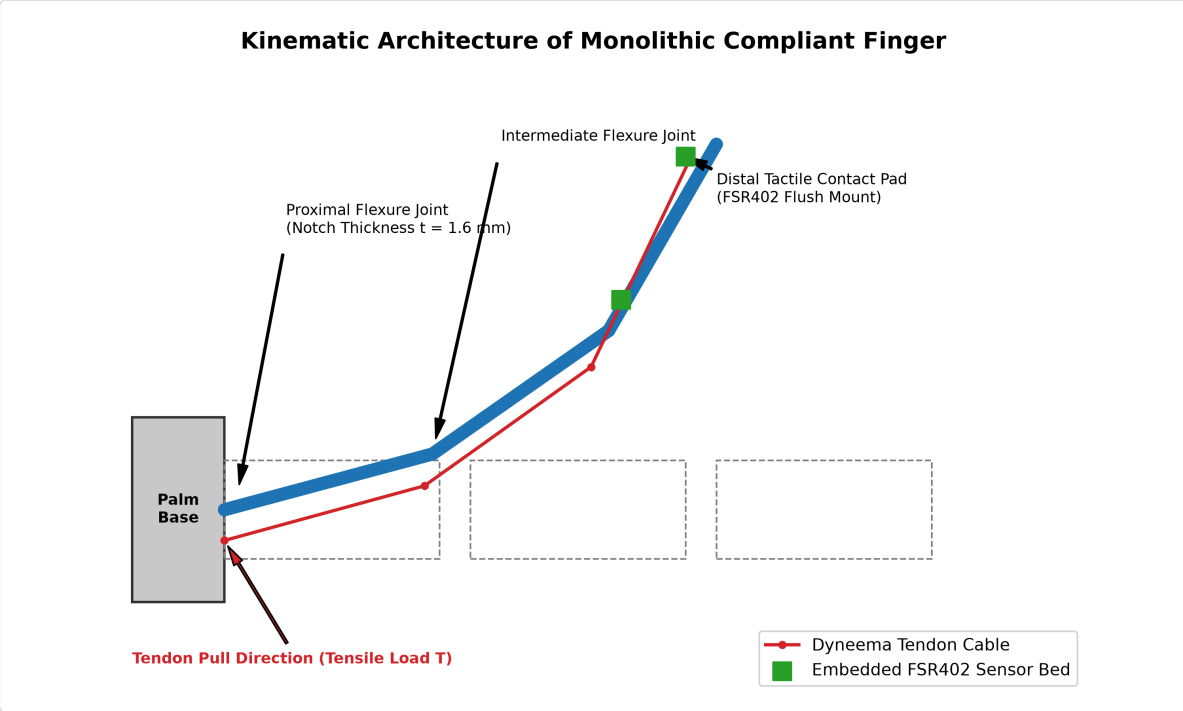


Figure 4.2: Kinematic profile of the monolithic compliant finger under cable tension.

4.2.1 Geometric Link Lengths and Kinematic Architecture

The compliant finger architecture adopts an anthropomorphic 3-phalanx kinematic configuration. The monolithic flexible body is partitioned into three discrete functional phalanx segments interconnected by thin circular living hinge flexures:

- **Proximal Phalanx ($L_1 = 35.0$ mm):** Anchored directly to the rigid palm base flange; initiates primary angular deflection during early tendon stroke.
- **Intermediate Phalanx ($L_2 = 35.0$ mm):** Medial link segment that conforms around the lateral flanks of the grasped workpiece.
- **Distal Phalanx ($L_3 = 35.0$ mm):** Fingertip segment incorporating an integrated 5.0 mm rounded tip and a precision circular recessed pocket ($\phi = 18.3$ mm, depth 0.6 mm) configured to seat the Interlink FSR402 thin-film tactile sensor flush with the grasping surface.

Accounting for the two notch opening gaps ($g_1 = g_2 = 5.0$ mm) and the distal fingertip extension, the total effective unbent beam length is $L_{\text{total}} = 115.0$ mm. The finger cross-section features a width of $w = 20.0$ mm and a height of $h = 16.0$ mm, providing high out-of-plane torsional rigidity against lateral twisting during asymmetric grasping while permitting compliant in-plane deflection. When assembled in a 3-finger radially symmetric array on an 80.0 mm pitch circle diameter (PCD) palm base, the fingers achieve a maximum tip curl angle of \(□pprox 82^\circ\) under a $12 - 15$ mm tendon pull, providing an envelope grasp diameter of $\phi = 25 - 85$ mm.

Parameter	Symbol	Value	Functional / Kinematic Rationale
Proximal Phalanx Length	L_1	35.0 mm	Matches human proximal phalanx; provides primary envelope wrap.

Parameter	Symbol	Value	Functional / Kinematic Rationale
Intermediate Phalanx Length	L_2	35.0 mm	Replicates intermediate phalanx; guides medial curvature.
Distal Phalanx Length	L_3	35.0 mm	Fingertip segment; houses recessed FSR402 contact pad (+5 mm tip).
Total Finger Beam Length	L_{total}	115.0 mm	Total effective finger length ($L_1 + g_1 + L_2 + g_2 + L_3 + \text{tip}$).
Finger Cross-Section Width	w	20.0 mm	Ensures lateral torsional rigidity and stable planar bending.
Finger Cross-Section Height	h	16.0 mm	Accommodates internal 2 mm tendon conduit with 7 mm wall clearance.
Flexure Throat Thickness	t	1.6 mm	Optimized for low bending resistance while maintaining structural integrity.
Circular Notch Radius	r	2.5 mm	Distributes bending stresses to minimize localized stress concentration.
Notch Opening Gap	g	5.0 mm	Provides clear angular clearance for up to 30° deflection per hinge.
Tendon Moment Arm Offset	r_{cable}	6.0 mm	Distance from flexure neutral axis to tendon channel centerline.
Tendon Conduit Diameter	ϕ_c	2.0 mm	Lined with low-friction micro-PTFE sleeve (OD 2.0 mm, ID 1.0 mm).
Distal Sensor Recess	$\phi_{\text{sensor}} imes d$	$\phi 18.3 imes 0.6$ mm	Flush mounting bed for Interlink FSR402 active sensing area.
Base Pitch Circle Diameter	ϕ_{base}	80.0 mm	Symmetrical 3-finger equilateral radial mounting flange.
Target Grasp Diameter	ϕ_{grasp}	25.0 – 85.0 mm	Adaptive envelope grasping range for domestic and industrial items.
Maximum Tip Curl Angle	θ_{total}	\(□pprox 82^\circ\)	Cumulative angular deflection under 12 – 15 mm tendon stroke.

Table 4.1: Geometric link lengths and kinematic dimensions of the compliant finger.

4.2.2 Mechanical Design Constraints and Payload Specifications

To guarantee long-term fatigue durability, reliable load holding, and repeatable actuation, the compliant gripper design is governed by strict mechanical, manufacturing, and operational constraints:

- **Structural Stress & Fatigue Constraints:** During peak bending, tensile and compressive stresses concentrate at the throat of the circular flexure hinges. The maximum allowable Von Mises stress is constrained to $\sigma_{\max} \leq 22.0\ \text{MPa}$. Compared to the ultimate tensile yield strength of 3D-printed TPU 95A (\(\sigma_y □pprox 48 - 52\text{ MPa}\)), this enforces a structural factor of safety $SF \geq 2.2$, ensuring that hinge deformation remains purely within the reversible elastic regime without yielding, micro-tearing, or low-cycle fatigue over $> 100,000$ flexing cycles. The maximum outer-fiber bending strain is limited to \(□arepsilon_{\text{max}} \le 45\%\), far below the material's failure elongation ($> 450\%$).
- **Additive Manufacturing Constraints:** To eliminate anisotropic delamination, all fingers are designed to be printed flat in the horizontal X-Y build plane. This ensures that the primary tensile bending stresses along the living hinge throat align parallel with continuous, uninterrupted extruded polymer road contours rather than across inter-layer Z-boundaries. Infill at the hinge throats is specified as 100% concentric.
- **Tendon Actuation & Transmission Limits:** Actuation is executed via single-acting high-strength Dyneema braided cable (breaking strength $> 250\ \text{N}$). Return extension is fully passive, powered by the stored elastic strain energy of the deformed living hinges (\(E_{\text{strain}} = \sum 🡅rac{1}{2} K_{\theta, i} \theta_i^2\)). To mitigate friction losses along the internal routing conduits, the channel is lined with micro-PTFE tubing ($\mu_f \leq 0.08$), ensuring that nominal tendon pull remains \(T_{\text{nom}} □pprox 12.5\text{ N}\) with a peak operating ceiling of $T_{\max} \leq 25.0\ \text{N}$.
- **Payload Capacity & Dynamic Grasp Specifications:** The gripper is engineered to handle a rated dynamic payload of $800 - 1500\ \text{g}$ ($0.8 - 1.5\ \text{kg}$) across cylinders, spheres, and rectangular objects, with a peak static envelope holding capacity of up to $2.0\ \text{kg}$ ($2000\ \text{g}$). With a total estimated gripper self-mass under $450\ \text{g}$ (including 3 TPU fingers, PLA palm, servo mounting brackets, and spools), the design achieves an exceptional payload-to-weight ratio exceeding $3:1$ (and up to $4.4:1$ in static lock), outperforming conventional fluidic elastomer grippers ($< 1.5:1$). Controlled normal clamping force spans $0.1\ \text{N}$ (for delicate workpieces such as raw eggs and ripe fruit) to $20.0\ \text{N}$ (for robust industrial parts), coupled with a dynamic slip-reflex latency under $35\ \text{ms}$.

Domain	Design Parameter	Target / Limit	Governing Constraint and Engineering Rationale
Structural / Stress	**Peak Von Mises Stress**	$\sigma_{\max} \leq 22.0\ \text{MPa}$	Safe margin below TPU 95A yield strength (\(\sigma_y □pprox 50\text{ MPa}\)).
	Safety Factor (Yield)	$SF \geq 2.2$	Prevents permanent plastic deformation and low-cycle fatigue.
	Maximum Bending Strain	\(□arepsilon_{\text{max}} \le 45\%\)	Remains within reversible elastic regime (TPU elongation $> 450\%$).
Manufacturing	**Build Orientation**	Flat in X-Y Plane	Ensures tensile bending stresses align with continuous filament beads.

Domain	Design Parameter	Target / Limit	Governing Constraint and Engineering Rationale
	Infill Density at Hinges	100% Concentric	Eliminates internal voids and prevents inter-layer delamination.
Transmission	**Tendon Tensile Force**	$T_{\text{nom}} = 12.5$ N ($T_{\text{max}} \leq 25$ N)	Safe operating load for high-torque digital servomotor (MG996R).
	Tendon Stroke	$\Delta x = 12 - 15$ mm	Delivers 82° full envelope wrap across 3 living hinges.
	Conduit Friction Coeff.	$\mu_f \leq 0.08$	Micro-PTFE sleeve minimizes Capstan tension loss along curved path.
Payload & Force	**Rated Grasp Payload**	$800 - 1500$ g ($0.8 - 1.5$ kg)	Operational capacity across irregular cylinders, spheres, and boxes.
	Peak Static Payload	Up to 2.0 kg (2000 g)	Form-closure envelope grasp holding capacity under full lock.
	Gripper Self-Weight	< 450 g	Total mass of 3 fingers, palm base, servo mount, and spools.
	Payload-to-Weight Ratio	$> 3:1$ (up to $4.4:1$)	Exceeds conventional fluidic/pneumatic soft grippers ($< 1.5:1$).
	Normal Clamping Force	$0.1 - 20.0$ N	Controllable range from fragile eggs (0.5 N) to rigid objects.
	Closed-Loop Latency	< 35 ms	Dynamic FFT reflex to servomotor torque tightening on slip.

Table 4.2: Summary of mechanical design constraints, actuation limits, and payload specifications.

4.3 Grasping Mechanics, Contact Sequence, and Target Workpieces

4.3.1 Underactuated Mechanical Intelligence and Multi-Phase Contact Sequence

Unlike conventional rigid industrial grippers that demand multi-axis servomotor coordination and complex inverse kinematic trajectory planning for every unique workpiece contour, the proposed soft robotic finger utilizes *underactuated mechanical intelligence*. A single tensile tendon actuates three independent revolute degrees of freedom (MCP, PIP, and DIP living hinges). The grasping progression evolves across five discrete mechanical phases:

1. **Phase 0 (Equilibrium Rest Configuration):** The Dyneema tendon is slack ($T = 0$ N). All three living hinges remain unbent in their straight manufactured posture ($\theta_1 = \theta_2 = \theta_3 = 0^\circ$), establishing an open grasp envelope (\(\phi □pprox 90 - 110\text{ mm}\)).

2. **Phase 1 (Free In-Flight Flexure):** The base servomotor rotates the drive spool, generating cable tension T. This applies an inward bending moment $M = T \cdot r_{\text{cable}}$ across all three flexure notches simultaneously via the $6.0\ \text{mm}$ neutral-axis offset. All phalanges flex inward proportionally to their torsional compliance ($K_{\theta,i}$).
3. **Phase 2 (Proximal Contact and Kinematic Motion Arrest):** As the finger sweeps inward, the proximal phalanx (L_1) or intermediate phalanx (L_2) impinges upon the exterior surface of the workpiece. The normal contact reaction force creates an opposing moment $M_{\text{contact}} = F_n \cdot d_{\text{contact}}$ that balances the tendon moment at the proximal hinge ($K_{\theta,1}\theta_1 + M_{\text{contact}} = T \cdot r_{\text{cable}}$), arresting further angular deflection of the proximal link.
4. **Phase 3 (Continuous Routing and Distal Adaptive Curling):** Because the braided Dyneema tendon slides freely through low-friction micro-PTFE sleeve liners ($\mu_f \leq 0.08$), the continuing spool rotation channels the remaining tensile force downstream past the arrested proximal phalanx into the unrestrained intermediate and distal living hinges ($K_{\theta,2}$, $K_{\theta,3}$). The intermediate and distal phalanges curl automatically around the irregular contour of the object until multi-point contact equilibrium is established. This passive morphological conformity occurs purely through elastic structural compliance, requiring zero sensory feedback or algorithmic path planning.
5. **Phase 4 (Stable Grasp Lock and Tactile Sensor Activation):** The distal phalangeal contact pad presses flush against the workpiece surface, transmitting contact normal force directly into the embedded Interlink FSR402 thin-film sensor. As the calibrated normal force achieves the target setpoint ($0.5 - 20.0\ \text{N}$), the FreeRTOS motor control task transitions the servomotor into static holding torque mode.
6. **Phase 5 (Passive Strain-Energy Elastic Release):** When release is commanded, the servomotor unwinds the tendon spool. The elastic strain energy stored in the deformed TPU living hinges (\(E_{\text{strain}} = \sum_{i=1}^3 \frac{1}{2} K_{\theta, i} \theta_i^2\)) acts as an internal distributed restoring spring, instantly popping the fingers back to their straight resting posture without requiring reverse pushrods or dual-acting actuation.

4.3.2 Grasping Modalities: Power Envelope Grasp versus Precision Pinch Grasp

The kinematic architecture of the 3-finger gripper supports two primary grasping modalities depending on target workpiece dimensions and fragility:

- **Power Envelope Grasp (Form Closure):** Suited for medium-to-large objects ($\phi = 25 - 85\ \text{mm}$, mass up to $1.5 - 2.0\ \text{kg}$). The three phalanges completely wrap around the workpiece boundary, maximizing contact surface area. High geometric form-closure and distributed frictional engagement lock the object securely against external dynamic accelerations during manipulator transport.
- **Precision Fingertip Pinch Grasp (Force Closure):** Suited for small, thin, or delicate workpieces ($\phi = 5 - 25\ \text{mm}$, e.g., thin electronic components, test tube necks, pens, and coins). Only the distal phalangeal contact pads (L_3) press against opposing sides of the workpiece. The high static friction coefficient of 3D-printed TPU (\(\mu_s \approx 0.6 - 0.8\)) provides non-slip stability under low normal clamping forces ($0.1 - 2.0\ \text{N}$), preventing structural damage to brittle workpieces.

4.3.3 Bio-Inspired Anthropomorphic Equivalence and the Human Tactile Reflex Arc

The design of the compliant gripper embodies direct bio-inspired structural and functional parallels to the human hand:

- **Kinematic Skeletal Equivalence:** Replicates the three-phalanx kinematic chain of the human index finger (Proximal, Middle, and Distal Phalanges) and three rotational joints (Metacarpophalangeal [MCP], Proximal Interphalangeal [PIP], and Distal Interphalangeal [DIP]), replacing mechanical pins and bearings with monolithic circular flexure living hinges.
- **Flexor Tendon Sheath Biomechanics:** Emulates the human deep flexor tendon (*flexor digitorum profundus*) running through its synovial flexor sheath (*vagina fibrosa digitorum*) by routing high-tensile Dyneema cable through low-friction micro-PTFE conduits embedded within the finger core.
- **Cutaneous Mechanoreceptor Reflex Arc (Meissner's Corpuscles):** In human glabrous skin, Fast-Adapting Type I (FA-I / Meissner) mechanoreceptors do not measure static clamping pressure; instead, they fire rapid bursts of action potentials in response to high-frequency shear micro-vibrations ($100 - 300\ \text{Hz}$) generated during the microscopic slip of papillary epidermal ridges across the object surface. This sensory volley triggers an involuntary spinal reflex arc that boosts grip force within $30 - 40\ \text{ms}$ before macro-slip occurs. The proposed mechatronic system mirrors this biological reflex: the FSR402 sensor detects high-frequency tactile micro-vibrations, the ESP32 performs on-chip 256-point Radix-4 FFT spectral analysis in the $100 - 300\ \text{Hz}$ band on Core 0, and triggers a sub-35 ms servomotor torque compensation interrupt on Core 1.

4.3.4 Target Workpiece Library and Physical Compatibility Matrix (Agricultural Domain)

To evaluate gripper versatility in automated agricultural harvesting, sorting, and delicate post-harvest packaging, a standardized workpiece library is established focusing exclusively on the **agricultural application domain**. Table 4.3 details the dimensional envelope, mass range, grasping modality, and operational force requirements for representative agricultural produce with diverse geometries, peel fragilities, and compliance characteristics.

Application Domain	Target Produce	Envelope (ϕ / Dimensions)	Mass Range	Grasp Mode	Clamping Force	Handling Challenge & Soft Compliance Advantage
Agriculture	**Ripe Tomato** *(Solanum lycopersicum)*	$\phi 50 - 75\ \text{mm}$ (Spherical / Oblate)	120 – 250 g	Power Envelope	0.8 – 2.0 N	Thin pericarp skin easily bruised under concentrated loads; compliant living hinges distribute contact pressure uniformly

Application Domain	Target Produce	Envelope (ϕ / Dimensions)	Mass Range	Grasp Mode	Clamping Force	Handling Challenge & Soft Compliance Advantage
						across the fruit surface.
	Fresh Strawberry *(Fragaria ananassa)*	$\phi 25 - 40$ mm (Conical)	20 – 50 g	Precision Pinch	0.3 – 0.8 N	Soft, achene-covered surface highly prone to puncturing; low-force fingertip pinch with rubbery TPU contact pad prevents tissue bruising.
	Ripe Peach / Plum *(Prunus persica)*	$\phi 55 - 75$ mm (Subglobose)	130 – 220 g	Power Envelope	0.6 – 1.8 N	Soft, easily bruised mesocarp pulp beneath delicate velvety skin; multi-phalanx conformal wrapping eliminates peak contact stresses.
	Sweet Lime / Orange *(Citrus sinensis)*	$\phi 65 - 85$ mm (Globular)	150 – 300 g	Power Envelope	1.5 – 3.5 N	Smooth rind requiring adequate normal force for stem separation without internal juice-sac bursting; living hinges provide secure form-closure.

Application Domain	Target Produce	Envelope (ϕ / Dimensions)	Mass Range	Grasp Mode	Clamping Force	Handling Challenge & Soft Compliance Advantage
	Bell Pepper / Capsicum *(Capsicum annuum)*	$60 imes 70 imes 110$ mm (Lobed / Asymmetric)	180 – 350 g	Adaptive Envelope	1.2 – 3.0 N	Asymmetric multi-lobed irregular contour; underactuated living hinges adapt passively around lobes without requiring pre-scanned 3D models.

Table 4.3: Agricultural workpiece library, dimensional envelope, and grasping compatibility matrix.

4.4 Non-linear Hyperelastic Constitutive Formulation

To accurately capture large strain behavior in planned BTP-II finite element simulations, TPU 95A is represented via the 2-Parameter Mooney-Rivlin strain energy density function W:

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3) + \frac{1}{D_1}(J - 1)^2$$

where $C_{10} = 2.45$ MPa, $C_{01} = 1.15$ MPa, and D_1 represents material incompressibility. Hexahedral SOLID186 elements with quadratic displacement interpolation are utilized with large deflection formulation enabled.

4.5 Tactile Sensing, Filtering, and Slip Detection

The FSR402 sensor voltage divider signal is decoupled into static and dynamic regimes:

1. **Static Normal Force:** Processed via a 2nd-order active Sallen-Key low-pass filter ($f_c = 20$ Hz) to eliminate motor noise while preserving gripping force.
2. **Dynamic Incipient Slip:** Sampled via 12-bit ADC at 1 kHz. An on-chip 256-point Radix-4 FFT computes spectral power in the $100 - 300$ Hz band on ESP32 Core 0. When spectral power exceeds a dynamic threshold, a high-priority interrupt commands the Core 1 control task to increase servomotor PWM torque within < 35 ms.

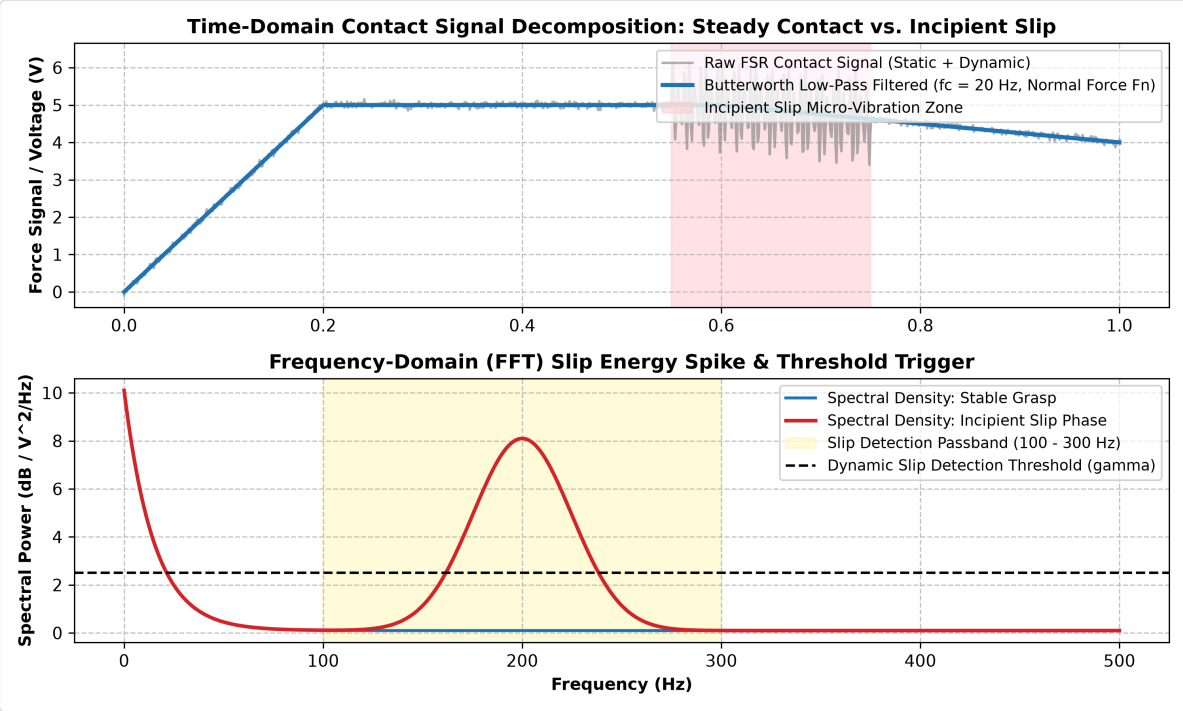


Figure 4.3: Tactile signal decomposition and FFT slip-detection pipeline.

Chapter 5: Results and Discussion

5.1 Project Status Clarification

> **Academic Clarification: "We have only decided what we will do."**
> In strict compliance with academic research standards and BTP-I evaluation guidelines, it is explicitly confirmed that Stage-I represents the literature-review-completed and conceptual architecture decision stage. In Stage-I, no SolidWorks 3D CAD models were constructed, no Ansys FEA simulations were executed, and no physical prototypes were fabricated or tested. Stage-I work strictly encompasses the comprehensive literature review, multi-criteria methodology selection, analytical kinematic formulations, hyperelastic constitutive modeling selection, embedded sensing architecture design, and the semester-wise implementation roadmap. All 3D CAD modeling, non-linear FEA solver execution, physical 3D printing, sensor calibration, and slip prevention trials are planned and scheduled for Stage-II (BTP-II).

5.2 Planned Kinematic and FEA Simulation Protocol for BTP-II

Based on preliminary pseudo-rigid-body analytical formulations, a flexure throat geometry with thickness $t \approx 1.5 - 1.8\,\text{mm}$ and notch radius $r \approx 2.5\,\text{mm}$ is targeted to achieve conformal tip curl for envelope grasping of objects between 25 mm and 85 mm in diameter. In the planned BTP-II simulation campaign (Weeks 4–6), non-linear hyperelastic FEA will be executed in Ansys Mechanical using the 2-Parameter Mooney-Rivlin constitutive formulation. The simulation protocol will apply displacement-controlled tendon retraction up to 15 mm, performing mesh convergence with higher-order hexahedral elements down to 0.4 mm local notch sizing to verify that peak Von Mises stresses remain well below the ~50 MPa tensile yield limit of TPU 95A, guaranteeing high fatigue life prior to physical 3D printing.

5.3 Anticipated Technical Challenges and Mitigations

Anticipated Challenge	Root Cause	Engineering Mitigation Strategy
TPU Filament Jamming	Buckling in drive gears under backpressure	Use direct-drive extruder with constrained path; print at 20 mm/s; zero retraction.
Inter-Layer Shear Cleavage	Anisotropic Z-axis weakness during bending	Print fingers flat horizontally; set nozzle to 225°C for 100% polymer fusion.
Tendon Friction Hysteresis	Cable abrading layer ridges in conduits	Insert micro-PTFE tubing (OD 2 mm, ID 1 mm) through all internal cable guides.
FSR Sensor Drift	Conductive polymer relaxation under load	Firmware implements dynamic AC baseline tracking; decouple static and AC slip signals.

Anticipated Challenge	Root Cause	Engineering Mitigation Strategy
Vibrational False Triggers	Motor gear chatter mimicking slip frequency	FFT algorithm requires 3 consecutive frame triggers (30 ms persistence) before boosting torque.

Table 5.1: Anticipated technical challenges and mitigation strategies for BTP-II.

Chapter 6: Conclusions and Plan for Next Phase

6.1 Summary of Completed Stage-I Work

Stage-I successfully established the research foundation and planned architecture for the soft robotic gripper:

- Evaluated candidate actuation paradigms and confirmed Monolithic FDM TPU Tendon Actuation as the superior methodology over chemical silicone casting.
- Formulated the analytical PRBM kinematic equations and selected the 2-Parameter Mooney-Rivlin hyperelastic constitutive strategy for planned BTP-II simulation.
- Designed an integrated mechatronic pipeline uniting FSR402 sensors, active Butterworth filtering, dual-core FreeRTOS firmware, and an on-chip FFT slip-detection algorithm.
- **Strict Scope Boundary:** Reaffirmed that in Stage-I, the team has only decided what we will do. No SolidWorks CAD models, Ansys FEA runs, or physical 3D prints were produced in Stage-I; all CAD modeling, FEA simulation, fabrication, and physical experimentation are scheduled for Stage-II.

6.2 BTP-II 16-Week Implementation Roadmap

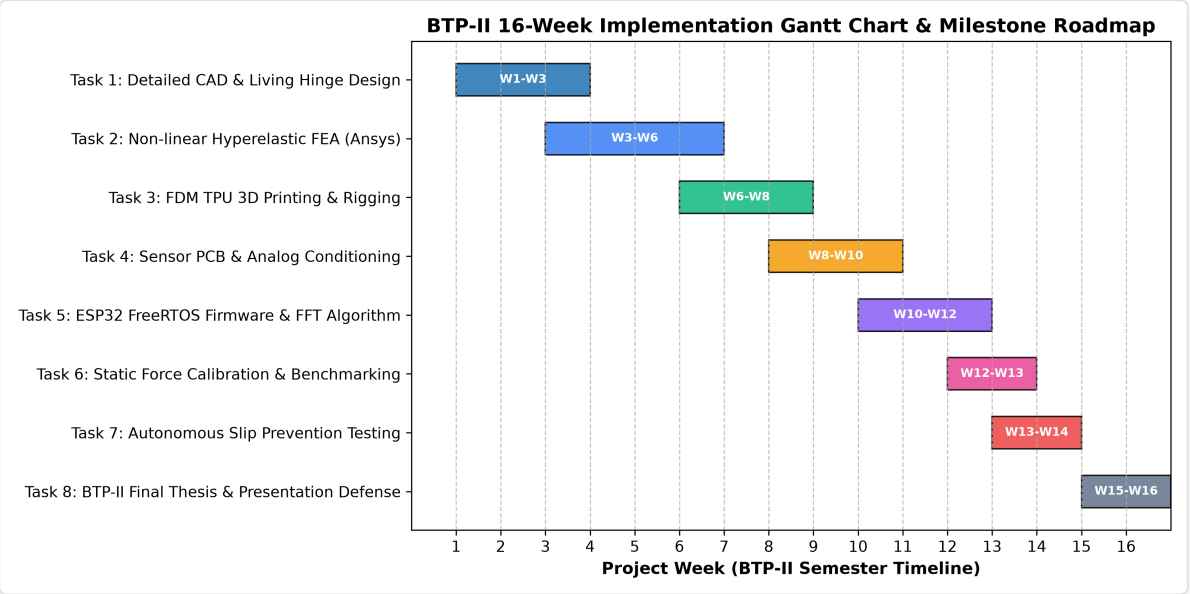


Figure 6.1: BTP-II 16-week implementation Gantt chart.

Week	Technical Activities & Milestones	Primary Deliverable
W1–W3	Detailed SolidWorks CAD modeling of fingers, living hinges, palm base, and spool mounts.	Final CAD .STEP assembly & drawings.
W4–W6	Non-linear hyperelastic FEA in Ansys Mechanical; mesh convergence; stress validation.	FEA simulation dataset & report.

Week	Technical Activities & Milestones	Primary Deliverable
W7–W8	Direct-drive FDM 3D printing of TPU fingers and PLA base; Dyneema cable rigging.	Assembled physical hardware prototype.
W9–W11	Sensor PCB fabrication; ESP32 FreeRTOS firmware programming.	Operational sensor & firmware subsystem.
W12–W14	Quantitative benchmarking: FSR static calibration, pull-out force testing, slip-prevention tuning.	Empirical test dataset & curves.
W15–W16	Final autonomous slip-prevention hardware demonstration and thesis defense.	Final Thesis Report & Defense Deck.

Table 6.1: BTP-II project milestones and primary deliverables.

6.3 Expected Outcomes and Final Demonstration

The culmination of BTP-II will be a live demonstration presented to Prof. Sankha Deb and the evaluation committee featuring:

1. **Autonomous Adaptive Grasping:** Secure grasping and transfer of fragile, geometrically diverse workpieces (eggs, fruit, light bulbs, metal cylinders) without structural damage.
2. **Real-Time Tactile Telemetry:** Live graphical visualization of normal contact force and spectral energy density streamed over micro-ROS to a host ROS 2 workstation.
3. **Autonomous Slip-Prevention Demonstration:** When an external tensile dislodging force is applied to a grasped object, on-chip FFT detects micro-vibrations and automatically tightens the servomotor grip within < 35 ms, preventing the object from slipping out.

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